

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [4]: data = {
    'Customer_ID': [101,102,103,104,105,106,107,108],
    'Age': [25,35,45,28,50,40,30,55],
    'Gender': ['Male','Female','Male','Female','Male','Female','Male','Female'],
    'Balance': [5000,12000,8000,15000,20000,10000,7000,25000],
    'Transactions': [10,25,15,30,40,20,12,50]
}
df = pd.DataFrame(data)
```

```
In [5]: print("First 5 Records:")
print(df.head())
```

First 5 Records:

|   | Customer_ID | Age | Gender | Balance | Transactions |
|---|-------------|-----|--------|---------|--------------|
| 0 | 101         | 25  | Male   | 5000    | 10           |
| 1 | 102         | 35  | Female | 12000   | 25           |
| 2 | 103         | 45  | Male   | 8000    | 15           |
| 3 | 104         | 28  | Female | 15000   | 30           |
| 4 | 105         | 50  | Male   | 20000   | 40           |

```
In [6]: print("\nDataset Info:")
print(df.info())
```

Dataset Info:

```
<class 'pandas.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Customer_ID      8 non-null     int64
1   Age              8 non-null     int64
2   Gender           8 non-null     str
3   Balance          8 non-null     int64
4   Transactions      8 non-null     int64
dtypes: int64(4), str(1)
memory usage: 452.0 bytes
None
```

```
In [7]: print("\nSummary Statistics:")
print(df.describe())
```

Summary Statistics:

|       | Customer_ID | Age      | Balance      | Transactions |
|-------|-------------|----------|--------------|--------------|
| count | 8.00000     | 8.00000  | 8.000000     | 8.000000     |
| mean  | 104.50000   | 38.50000 | 12750.000000 | 25.250000    |
| std   | 2.44949     | 10.86278 | 6881.652625  | 14.109267    |
| min   | 101.00000   | 25.00000 | 5000.000000  | 10.000000    |
| 25%   | 102.75000   | 29.50000 | 7750.000000  | 14.250000    |
| 50%   | 104.50000   | 37.50000 | 11000.000000 | 22.500000    |
| 75%   | 106.25000   | 46.25000 | 16250.000000 | 32.500000    |
| max   | 108.00000   | 55.00000 | 25000.000000 | 50.000000    |

```
In [8]: print("\nAverage Balance:", df['Balance'].mean())
```

Average Balance: 12750.0

```
In [9]: print("Average Transactions:", df['Transactions'].mean())
```

Average Transactions: 25.25

```
In [10]: print("\nGender Distribution:")  
print(df['Gender'].value_counts())
```

Gender Distribution:

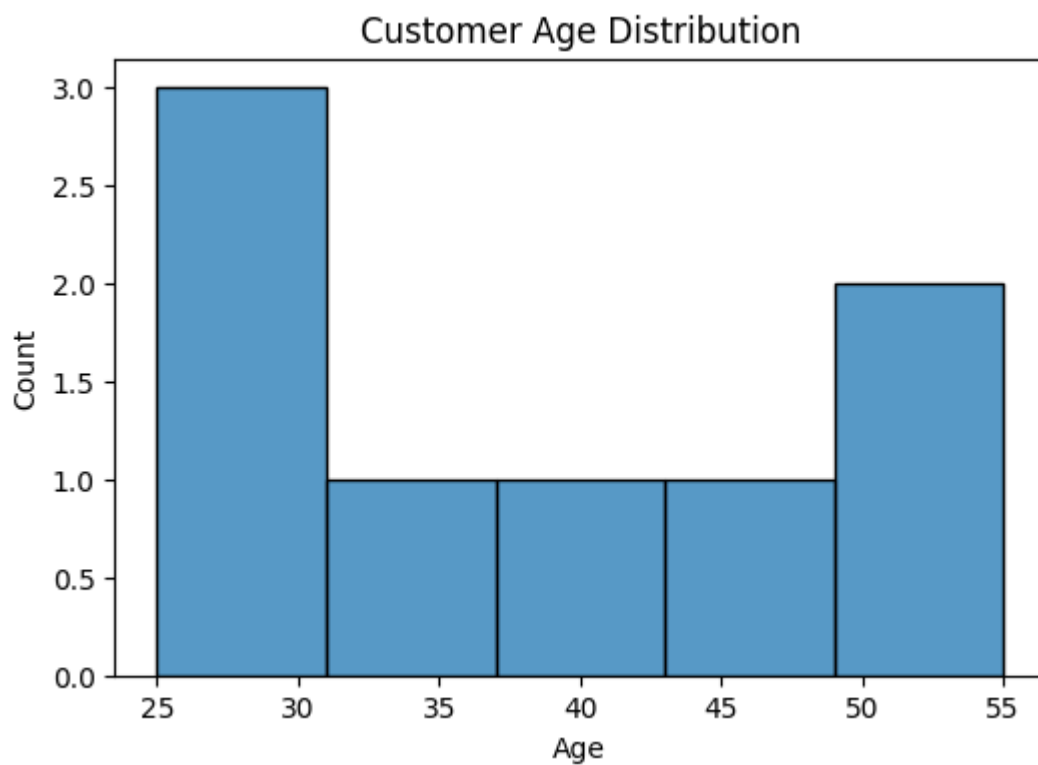
Gender

Male 4

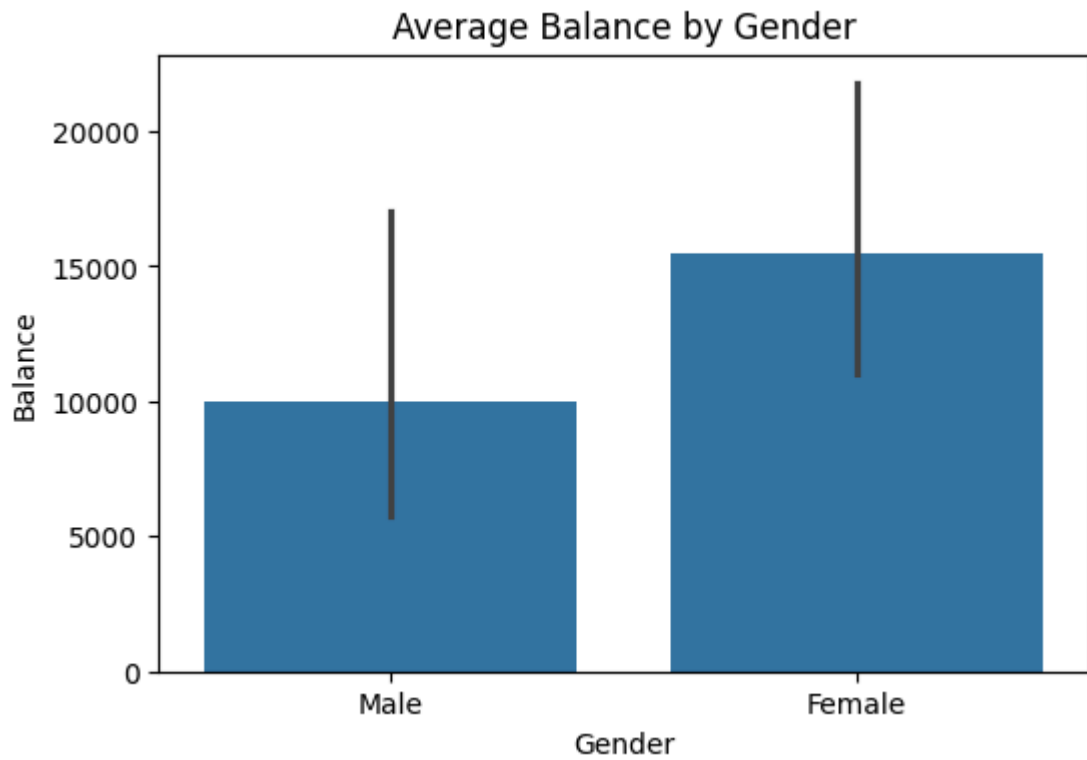
Female 4

Name: count, dtype: int64

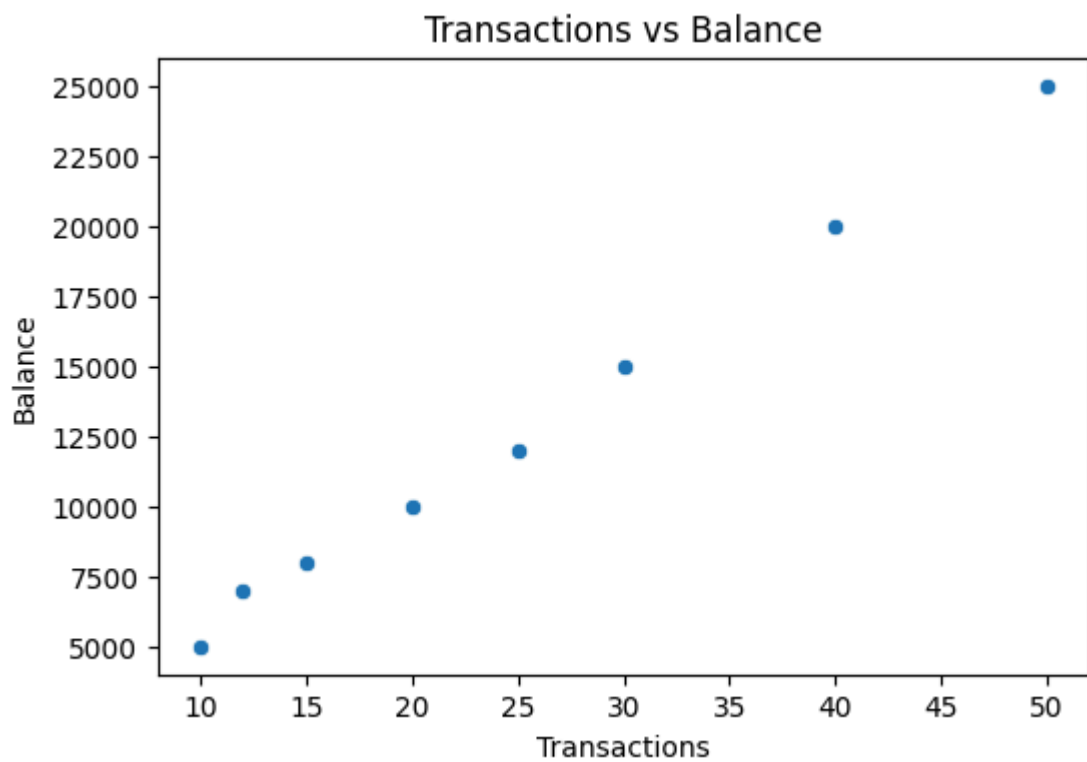
```
In [11]: plt.figure(figsize=(6,4))  
sns.histplot(df['Age'], bins=5)  
plt.title("Customer Age Distribution")  
plt.show()
```



```
In [12]: plt.figure(figsize=(6,4))  
sns.barplot(x='Gender', y='Balance', data=df)  
plt.title("Average Balance by Gender")  
plt.show()
```



```
In [13]: plt.figure(figsize=(6,4))
sns.scatterplot(x='Transactions', y='Balance', data=df)
plt.title("Transactions vs Balance")
plt.show()
```



```
In [ ]:
```